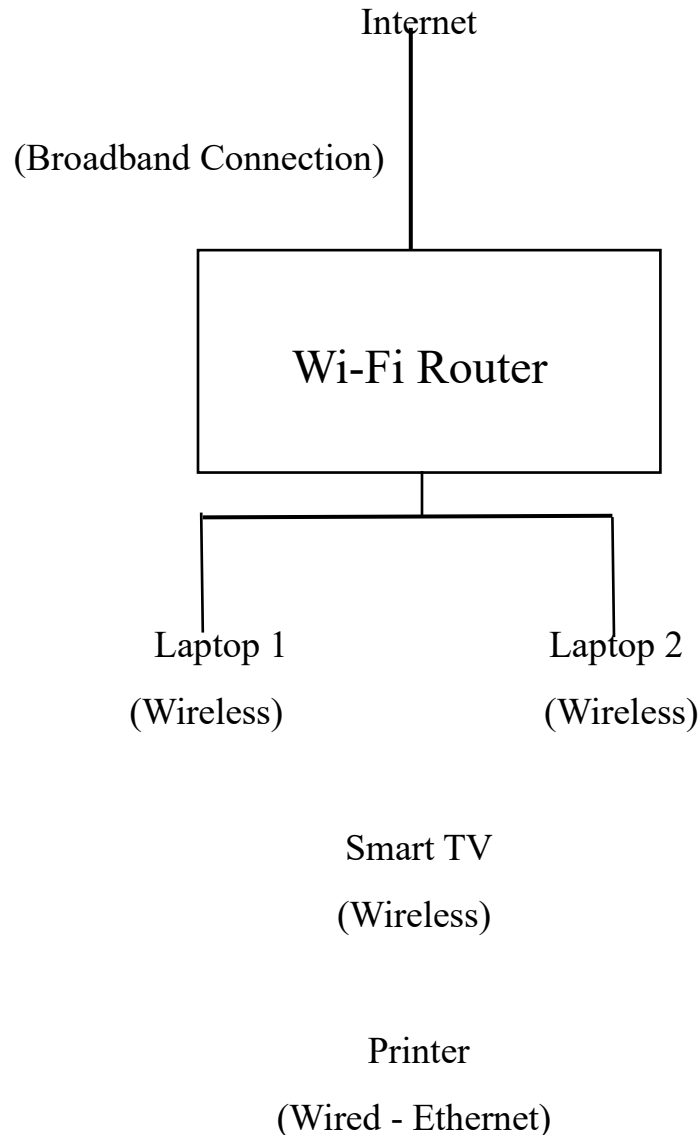


BUILD A SOHO NETWORK

1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



**2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.**

Hardware	Suggested Brand/Model	Purpose
Broadband Connection	ISP (JioFiber, Airtel Xstream, BSNL FTTH)	Provides Internet connection.
<u>TP-Link Archer C6 AC1200 Gigabit Router</u>	TP-Link	Connects all wired and wireless devices.
Laptop 1	Dell Inspiron 15 / HP 15	Used for office or personal work.
Laptop 2	Lenovo IdeaPad Slim 3	Used for office or personal work.
Smart TV	Samsung 43" Smart TV / LG Smart TV	Used for streaming YouTube, Netflix, etc.
<u>HP LaserJet 1020 Plus Printer</u>	HP	Prints documents.
<u>D-Link CAT6 Ethernet Patch Cord</u>	D-Link	Connects the printer to the router (wired).
<u>Netgear GS305P 5-Port Gigabit Ethernet Switch</u> (Optional)	Netgear	Adds extra Ethernet ports if needed.
<u>TP-Link RE205 AC750 Wi-Fi Range Extender</u> (Optional)	TP-Link	Extends Wi-Fi coverage if the signal is weak.

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Steps to Configure a Wi-Fi Router for a SOHO Network

Step 1: Connect the router to the broadband modem/Internet connection and turn it on.

Step 2: Connect your laptop to the router using Wi-Fi or an Ethernet cable.

Step 3: Open a web browser and enter the router's IP address (usually 192.168.0.1 or 192.168.1.1).

Step 4: Log in using the default username and password (printed on the router or in the manual).

Step 5: Go to Wireless Settings and set a unique SSID (Wi-Fi Name), for example HomeOffice_WiFi.

Step 6: Select WPA2-PSK (or WPA2/WPA3 if available) as the security mode.

Step 7: Create a strong Wi-Fi password (at least 8 characters with letters, numbers, and symbols).

Step 8: Change the default admin password to a new secure password.

Step 9: Click Save/Apply and restart the router if prompted.

Step 10: Connect your laptops, smart TV, and printer to the new Wi-Fi using the new SSID and password.

- SSID: HomeOffice_WiFi
- Security: WPA2-PSK
- Admin Password: Changed from the default to a secure password
- All devices: Successfully connected to the SOHO network.

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Device	MAC Address (Example)	IP Assignment	IP Address
Wi-Fi Router	00:1A:2B:3C:4D:01	Static	192.168.1.1
Laptop 1	00:1A:2B:3C:4D:11	DHCP	192.168.1.100
Laptop 2	00:1A:2B:3C:4D:12	DHCP	192.168.1.101
Printer	00:1A:2B:3C:4D:21	Static	192.168.1.10
Smart TV	00:1A:2B:3C:4D:31	Static	192.168.1.20

- DHCP Devices: Laptop 1 and Laptop 2 automatically receive IP addresses from the router.
- Static Devices: The printer and Smart TV use fixed IP addresses so they are always available at the same address.
- Router IP: 192.168.1.1 (Default Gateway)
- Subnet Mask: 255.255.255.0

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

1. Wi-Fi Not Connected

Cause: The laptop is not connected to the correct Wi-Fi network.

Fix:

- Check if Wi-Fi is turned on.
- Connect to the correct SSID.

- Enter the correct Wi-Fi password.

2. IP Address Problem

Cause: The laptop did not receive a valid IP address from the router.

Fix:

- Run `ipconfig` to check the IP address.
- Run `ipconfig /release` and `ipconfig /renew`.
- Restart the router if needed.

3. Internet or Router Issue

Cause: The router has no Internet connection or is not working properly.

Fix:

- Check the router's Internet/WAN light.
- Restart the router.
- Test the Internet on another device. If all devices fail, contact the ISP.

4. DNS Problem (Optional)

Cause: The Internet is connected, but websites do not open.

Fix:

- Try opening another website.
- Change the DNS server to 8.8.8.8 or 1.1.1.1.
- Flush the DNS cache using `ipconfig /flushdns`